

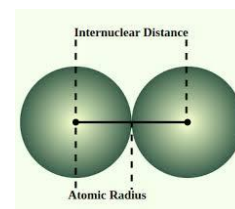
ATOMIC STRUCTURE

2.1: Atomic Radius

Average distance of nucleus from the outermost electron is called atomic radius.

OR

An atomic radius is half the distance between adjacent atoms of the same element in a molecule.



Periodic trend

Across the period

As you move from left to right, the atomic number increases, which increases the effective nuclear charge. The more positive protons in the nucleus pull on the negative electrons, bringing them closer in. This increase in attractive forces reduces the atomic radius.

From left to right the shielding effect remain constant. However, in transition metal series the shielding effect increased as the electrons are configured in the inner shell d-orbitals.

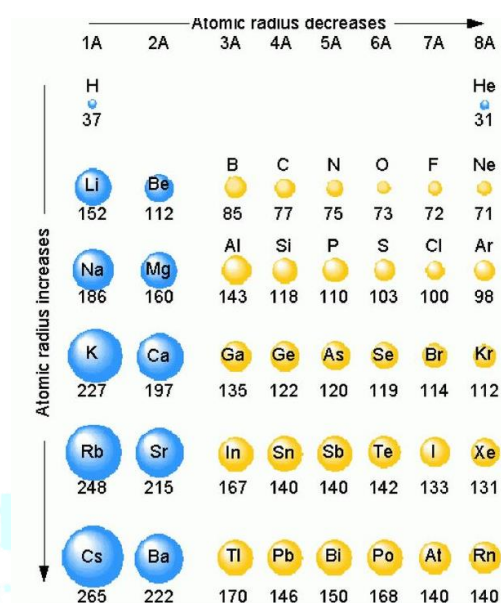
Down a group

As you move down a group, new energy levels are needed to hold the electrons, providing extra "layers" that make the atom larger.

Reason

No of shells are increased.

Shielding effect increased due to inner shell electrons.



Atomic Radius Questions

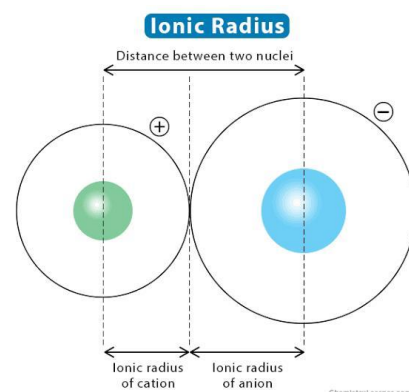
- Why atomic size of sodium is less than potassium?
- Why radius of atom is increased down the group?
- Which of the following has more atomic radius?
A) Li Or Na. B) Be Or Mg. C) O Or N
- Arrange the elements in order of increasing atomic radius.
A) Li, B, C, Be. B) Al, Na, Si, Mg

2.2: Ionic Radius

“The average distance from the nucleus to the outermost electron in ion is called ionic radius”

OR

“The half of the distance between the nuclei of two ions is called ionic radius.”

$$\text{Na}^+ < \text{Na}$$
$$\text{O}^{2-} \rightarrow \text{O}$$


Increased in shielding effect

¹H 1.38 1.06 0.00 —																	²He — 0.62 0.53 0.00						
³Li 3.33 2.69 0.39 0.35	⁴Be — 1.79 1.59 0.29	atomic number → ⁶C ← atomic symbol radius of C⁻ → 1.91 ← radius of C radius of C⁺ → 1.25 ← radius of C²⁺ 0.95 ←																⁵B 2.78 1.86 1.23 1.14	⁶C 1.91 1.49 1.25 0.95	⁷N — 1.24 1.08 0.96	⁸O 1.29 1.13 0.94 0.85	⁹F 1.13 0.98 0.88 0.76	¹⁰Ne — 0.87 0.78 0.72
¹¹Na 3.57 2.91 0.71 0.66	¹²Mg — 2.18 1.06 0.60																	¹³Al 3.63 2.60 1.65 1.54	¹⁴Si 2.58 2.15 1.87 1.36	¹⁵P 2.21 1.83 1.64 1.50	¹⁶S 1.84 1.68 1.46 1.35	¹⁷Cl 1.57 1.46 1.37 1.23	¹⁸Ar — 1.29 1.22 1.16
¹⁹K 4.36 3.59 1.11 1.06	²⁰Ca — 2.81 2.55 0.98	²¹Sc 2.57	²²Ti 2.42	²³V 2.30 1.94 1.46	²⁴Cr 2.38 1.72 1.41	²⁵Mn 2.11 1.97 1.30	²⁶Fe 2.09 1.91 1.28	²⁷Co 2.05 1.48 1.21	²⁸Ni 2.00 1.45 1.17	²⁹Cu 2.77 2.24 1.38 1.14	³⁰Zn 1.93 1.75 1.09	³¹Ga 3.70 2.64 1.58 1.49	³²Ge 2.64 2.21 1.94 1.38	³³As 2.33 1.94 1.76 1.62	³⁴Se 1.97 1.82 1.60 1.50	³⁵Br 1.72 1.62 1.53 1.39	³⁶Kr — 1.47 1.40 1.34						
³⁷Rb 4.65 3.84 1.29 1.25	³⁸Sr — 3.08 2.80 1.16	³⁹Y 2.94	⁴⁰Zr 2.55 2.41 1.89	⁴¹Nb 2.58 2.10 1.75	⁴²Mo 2.43 1.88 1.65	⁴³Tc 2.20 2.10 1.53	⁴⁴Ru 2.43 1.67 1.48	⁴⁵Rh 2.45 1.57 1.39	⁴⁶Pd 1.81 1.51 1.33	⁴⁷Ag 2.99 2.46 1.42 1.28	⁴⁸Cd 2.14 1.96 1.22	⁴⁹In 3.71 2.81 1.80 1.70	⁵⁰Sn 2.63 2.44 2.18 1.60	⁵¹Sb 2.58 2.17 2.00 1.86	⁵²Te 2.22 2.07 1.85 1.74	⁵³I 1.96 1.87 1.78 1.64	⁵⁴Xe — 1.71 1.64 1.55						

Arrange the following in increase order of their radius.

Na, Cl, Na⁺, Cl⁻

Which of the following has bigger size and why?

A) Mg, Mg²⁺ b) O or O²⁻

2.3: Quantum Number

Principal Quantum Number (n)

- Orbit, shell, circular path
- This quantum number is the expression of Bohr's shell in quantum form.
- It is represented by n, which has possible values n=1,2,3...
- The principal quantum number also depicts the period number in the periodic table, because each period starts with the new shell. This number represents the Bohr's shell.
- If n is greater the energy of the shell is greater so is its size.
- Principal quantum Number arose from the emission spectrum of hydrogen which was first of all explained by Bohr and Max Planck.
- Each spectral line in the emission spectrum is explained by the possible jumps within shells of H atoms.
- **Applications:**
- Radius of an atom = $0.529A \times n^2$
- Energy of the orbit = $1313.315/n^2$ KJ/mol
- Number of e in an orbit = $2n^2$

Azimuthal Quantum Number (l)

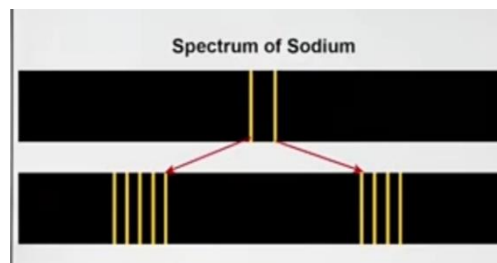
- The azimuthal quantum number which is denoted by l describes the shapes of orbitals of electronic shells.
- An orbital is a space around the nucleus where the probability of finding the electron is maximum. One shell may be comprising of more than one subshell or orbitals.
- Azimuthal quantum number determines the orbital's angular momentum of the electron and specifies the subshell within a given principal energy level (n). The values of l range from 0 to n-1.
- Possible values: l=0,1,2,3...
- The presence of subshells is related to the fine spectral lines in the spectrum. Number of subshells in a shell depends upon the shell number, bigger the shell, more are the number of subshells in it and more electrons can be accommodated in it.
- Give the number of subshells in a shell.

Magnetic Quantum Number (m)

The splitting of spectral line into more closely spaced line under the influence of strong magnetic field can be explained on the basis of magnetic quantum number.

- Also called as the orientation quantum number.
- Bohr's atomic model cannot explain the Zeeman effect. It the defect of Bohr's atomic model
- Zeeman effect:

- When excited atom placed into the magnetic field the spectral line further split up into more closely spaced line. These is called Zeeman effect. These closely spaced lines gives us the concept about the orbitals.
- Values of m :
 $m=0, \pm 1, \pm 2, \pm 3, \dots$
- Value of m tell us about:
 - ✓ Degeneracy of orbitals in space
 - ✓ Different ways in which subshells can be arranged into x,y and z axis in the presence of magnetic field
 - ✓ Different space orientation of a subshell



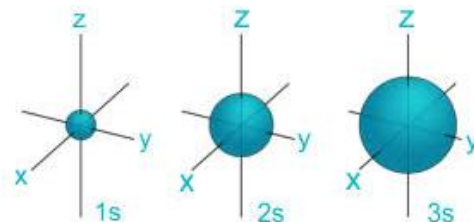
Number of orbitals in a subshell:

By using the formula $n=2l+1$ we can find the number of orbitals in a subshell

a) S-subshell:

For s- subshell $l=0$, m has one value which is $m=0$

Shape is spherically symmetrical



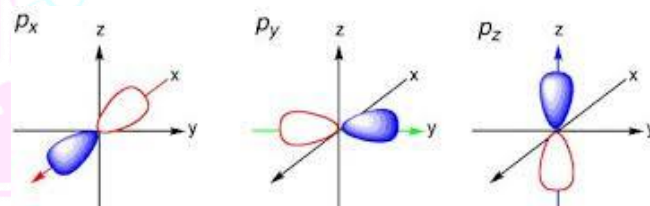
b) p-orbitals:

For p-subshell, $l=1$, so m has three values which are

$$m = -1, 0, +1$$

(P_x) (P_z) (P_y)

Dumbbell shape

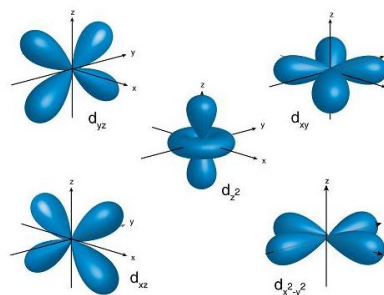


c) d-subshell:

- For d-subshell, $l=2$ so m has five values.
 $m = -2, -1, 0, +1, +2$
- So, d has five orbitals or five space orientation.
- In d_{xy} , d_{yz} and d_{xz} the lobes lie between the axis.
- In $d_{x^2-y^2}$ the lobes lie on the axis. In d_{z^2} both lobes lie on the z-axis axis and make a collar. (collar rotation, electronic rotation)

No of electrons in a orbital:

- S, p_x, p_y, p_z , d_{xy}, d_{yz}, d_{xz} , $d_{x^2-y^2}$, d_{z^2}
- To find the number of orbitals in an orbit
- we use the formula n^2
- For first orbit $n=1$, so, only one orbital
- Which is s-orbital.
- For 2nd orbit $n=2$
- Number of orbitals are four. S, p_x , p_y and p_z .



- How many orbitals are in 3rd orbit?
- For principal quantum number
- **Number of electrons in an orbit=2n²**
- For Azimuthal Quantum Number
- **Number of electron in subshell= 2(2l+1)**
- For Magnetic Quantum Number
- Number of electrons in an orbital=2(maximum)

Spin Quantum Number (s)

1. Electron Occupancy in Orbitals

- **Maximum of two electrons** can reside in a single orbital.
- These electrons must have **different identifications**, which includes having **opposite spins**.

2. Role of Spin in Electrons

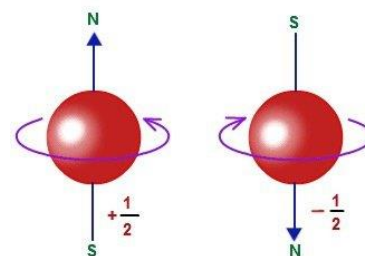
- Spin is a fundamental property of electrons.
- Electrons in the **same orbital** must have **opposite spins** to **cancel each other's spin magnetic field**.

3. Discovery of Electron Spin

- In **1925**, **Goudsmit** and **Uhlenbeck** proposed the concept of **electron spin**.
- They suggested that electrons **spin about their own axis**, like a self-rotation.
- This spinning creates a **magnetic field** around the electron (called **spin magnetic field**).

4. Spin and Magnetic Field Cancellation

- To share an orbital, the two electrons must **neutralize their magnetic fields**.
- This is achieved when one spins **clockwise** and the other **anticlockwise**.



5. Representation of Spin Quantum Number

- Clockwise spin: $+\frac{1}{2}$, symbol: $\downarrow$
- Anticlockwise spin: $-\frac{1}{2}$, symbol: $\uparrow$

2.4: Electronic distribution

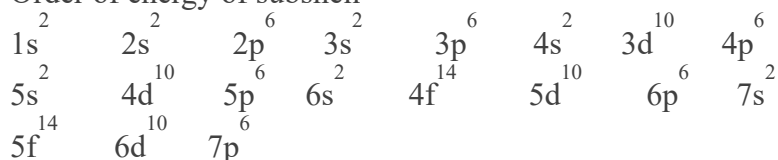
“The distribution of electrons in shell, subshell and orbitals of an atom is called atomic distribution. An orbital can accommodate maximum of two electrons.”

Number of electrons in a shell= $2n^2$

Number of electrons in a subshell are s=2, p=6, d=10, f=14

(n+l) rule

Order of energy of subshell



Larger n+l value has more energy

For 1s, $1+0=1$ For 2s, $2+0=2$ For 2p, $2+1=3$

If n+l value is same for two subshells then the subshell With large principal quantum number has more energy value.

For 2p, $2+1=3$ For 3s, $3+0=3$

So, 3s has more energy than 2p.

Auf Bau principal:

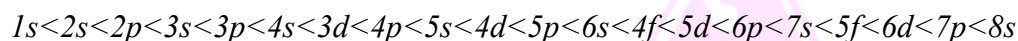
Auf Bau is German language word means “building together”

Tells about distribution of electrons in subshell.

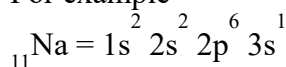
According to the Auf Bau principal

“Electrons are distributed in orbitals or subshells according their increasing order of energy.”

The increasing order of energy level is following based upon the n+l rule



For example



The orbitals which has less energy is filled first.

Concept assessment 2.41. In which orbital of the atom the 11th electron will be configured?

Answer: 3s orbital

2. Place the orbital in increasing energy order 5s, 4p, 4s, 3d?

Answer: $4s < 3d < 4p < 5s$ **Concept assessment 2.5**

Which of the following orbitals has greater energy?

1. 5f or 6p

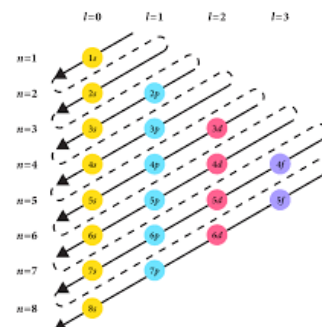
Answer: As the n+l value for 5f is greater than 6p so it has greater energy than 6p

2. 5s or 4d

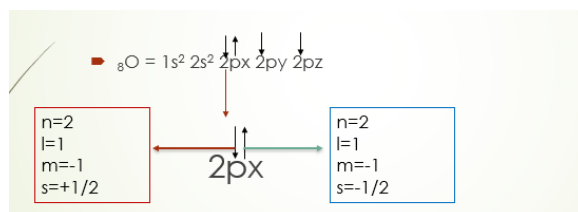
Answer: As n+l value for 4d is greater than 5s so it has more energy than 5s.

Pauli exclusion principal:

It can be stated as;

“No two electrons in atom can have the same value of all the four-quantum number.”

Two electrons in orbitals have same value of n , l , m but its spin quantum number will be different.
For example



So, one electron has clockwise rotation ($+1/2$) and other has anticlockwise rotation ($-1/2$).

Hund's Rule:

According to this rule

“If degenerate orbitals are available

- Electron tends to reside separately with same spin
- Don't pair up in same orbital with opposite spin

Rule tell about the

Degeneracy rule

Degenerate orbitals

Valency of an element

For example

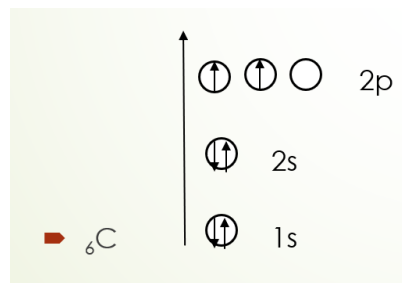
Energy of the orbitals is also related with the number of electrons present in it.

For example, half-filled and completely filled orbitals are more stable with less energy.

Topic related questions

- ✓ Write the subshells in increasing order of energy by explaining with $n+l$ rule?

1. 3d, 4s, 4p
2. 2p, 2s, 3d
3. 4s, 4p, 4d, 4f



Concept Assessment 2.3

1. Role of principal quantum number in describing the Energy levels of electron within an atom.

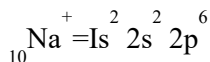
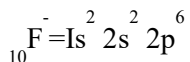
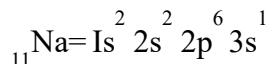
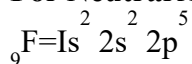
Answer: The principal quantum number, n , defines the allowed energy levels, with higher values of n corresponding to higher energy levels. The value of n also determines the number of available subshells and orbitals for the electron, resulting in more complex electron configurations as n increases.

2. When the principal quantum number (n) is 3 how many different possible values can the azimuthal quantum number (l) take?

Answer: When the principal quantum number (n) is 3, the azimuthal quantum number (l) can take 3 different possible values. $l = 0, 1, 2, \dots, (n-1)$ When $n = 3$, the values of l can be: $l = 0, 1, 2$. This means that for $n = 3$, the possible values of the azimuthal quantum number (l) are 0, 1, and 2.

3. What is the significance of the magnetic quantum number (m) in relation to the arrangement of electrons within an atom?

The magnetic quantum number (m) determines the orientation of an electron's orbital angular momentum in an atom. Each m value corresponds to a distinct atomic orbital, which affects the electron's probability distribution. This influences the arrangement of electrons in an atom's configuration.

Electronic configuration of Neutral atom and Ion**For Neutral Atom:****For Fluorine anion****For sodium cation**

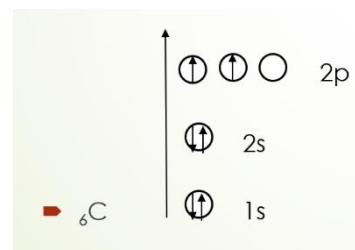
Electronic configuration for $\text{H}^+ = 1\text{s}^0$

Electronic configuration in term of energy of electrons and inter-electron repulsion

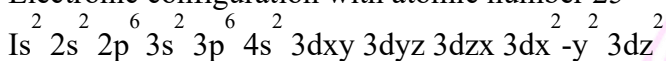
As we know that each orbital accommodates maximum of 2 electrons. Three two electrons can repel each other in same orbital due to the same charge. Due to this reason when different degenerate orbitals are available then electrons are allowed to filled in different orbitals of same subshell. (Hund's Rule)

Spin pair repulsion:

Electron with same spin repel each other this is called as the spin pair repulsion. Electron occupy separate orbitals in same orbitals to minimize the repulsion (Hund's Rule)



Electronic configuration with atomic number 25



$n=3$

$l=2$

$m=-2, -1, 0, +1, +2$

$S=+1/2$

Concept assessment 2.5

1. Write the electronic configuration of following elements representing all the four quantum numbers.

a) ${}_{21}\text{Sc}$ b) ${}_{24}\text{Cr}$ c) ${}_{25}\text{Mn}$ d) ${}_{30}\text{Zn}$

2. Write electronic configuration of following ions.

A) Na^+ b) C^{4-} c) Ca^{2+}

Shapes of Orbitals

Name of orbital	Shape name	Shape
s-orbital	Spherical shaped	
p-orbital	Dumbbell shaped	

d-orbital	Cloverleaf shaped	
f-orbital	Complicated shaped	

2.5: Free radicals:

“An atom or group of atoms having unpaired electron in valance shell is called free radical.”

It is represented by a (.) dot.

For example

Cl has 7 electrons in its valance shell so it forms free radical

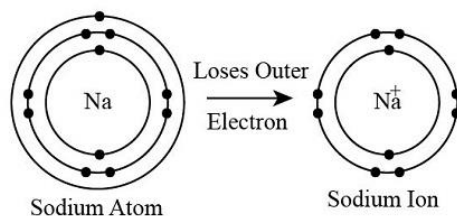
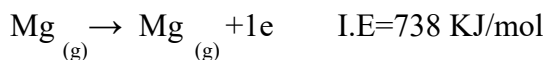
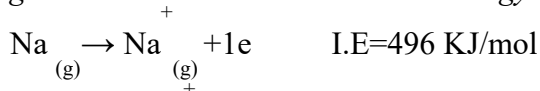


2.6: Ionization Energy:

Ionization enthalpy or ionization potential

It can be defined as

“The minimum amount of energy required to remove the electron form the valance shell of an isolated gaseous atom is called ionization energy”



- Some atoms have 2 or more electrons in the valance Shell. For example, for example alkaline earth elements such as Mg or Sr
- Removing the 1st electron (first ionization energy) will always take less energy than the energy necessary to Remove the 2nd electron (second ionization energy).
- This trend continues. The energy necessary to remove a 3rd electron is greater than the energy necessary to remove the 2nd electron.

Important points related to ionization energy

Ionization energy is directly proportional to the attraction between the nucleus and valance electron.
I.E always has positive value.

Unit if I.E are KJ/mol, Kcal/mol, eV/atom.

It is a kind thermodynamic property.

I.E is directly related to the stability of a substance. (i.e high I.E difficult to remove the electron)

Factor affecting the I.E

1. Nuclear Charge:

As we move form left to right in the periodic table;

Atomic number is increased

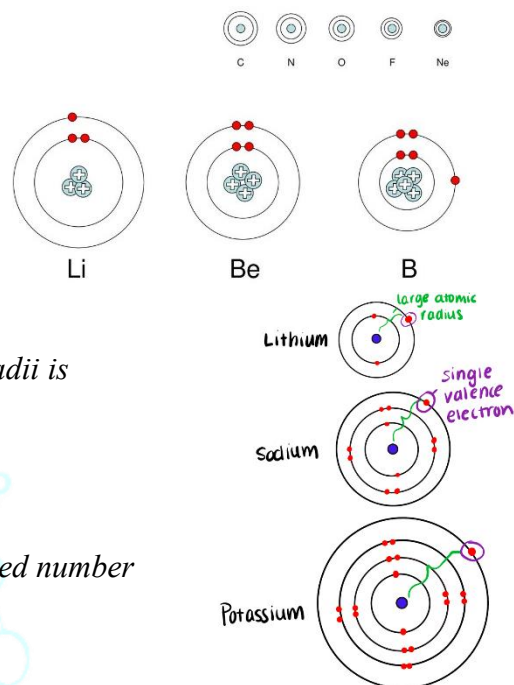
Nuclear charge is increased

The force of attraction of nucleus on the valence electron is increased.

It became difficult to remove the outermost electrons.

More energy is required

So, ionization energy is increased.



2. Atomic Radii

As we move for left to right in period of periodic table atomic radii is decreased.

Valence shell come near to the nucleus.

Attraction between nucleus and valence electron increased.

More energy required to remove the valance electron.

So, ionization energy is increased.

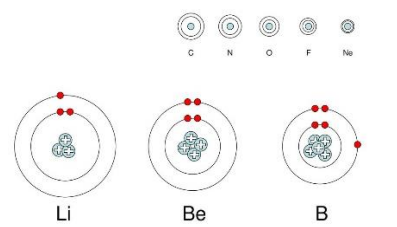
In group from top to bottom atomic radii is increased to increased number of shell.

Distance between valance shell and nucleus is increased.

Force of attraction of nucleus on valance electron is decreased.

Become easy to remove the electrons.

I.E is decreased.



3. Shielding effect:

From left to right shielding effect remain constant so it has no effect on I.E

From top to bottom shielding effect is increased due to increased number of shell.

The more inner shell electron decreases the force of attraction of nucleus on the valence shell electron.

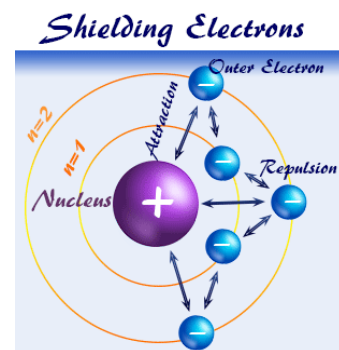
It became easy to remove the electron.

Less energy is required to remove the electrons.

So, from top to bottom I.E is decreased.

4. Spin pair repulsion:

When electrons are paired in the half-filled p orbitals it causes the electron pair repulsion which cause the energy of electron to increase.

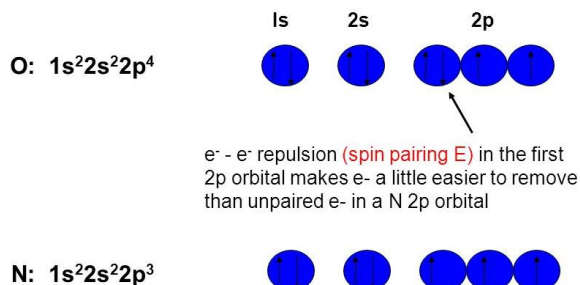


The increase energy makes the electron easier to remove.

So, less energy required to remove the electron.

For Example,

I.E of O is less than N



Periodic trend

Trend in Period:

The ionization energy increases across a period due to the decreasing atomic size and increasing effective nuclear charge. The shielding effect remains constant in the period so it has no effect on the I.E. However, spin pair repulsion can cause some variations in the trend. The electron pairing in the half filled p orbitals cause the ionization energy to decrease.

Trend in group:

In group from top to bottom ionization energy is increased because we more form top to bottom the atomic size and shielding effect is increased which decrease the force of attraction of nucleus on the valance electron and it become easy to remove the valance shell electron. So, ionization energy is decreased.

Question: Arrange the elements in order of increasing ionization energy.

a) Na, Li, K b) Be, Li, B c) C, N, O



Deduction of position of an element in the periodic table

Trend in ionization energy

- ✓ Left to right increase

- ✓ Top to bottom decrease

Periodic table organization

Elements with comparable energy grouped together.

Comparing the Ionization Energy

To find the position of element in the periodic table, the ionization energy data is compared.

Example

In period suppose "X" is an element with 1251 KJ/mol. It should be present between S (1000 KJ/mol) and Ar (1520 KJ/mol). So, "X" should be Cl

In group suppose "y" is an element with I.E value 738 KJ/mol. It is present between Be (899 KJ/mol) and Ca (890 KJ/mol). So, "y" should be Mg

Don't always the same trend.

Electronic configuration and position in periodic table

Comparing 1st, 2nd and 3rd ionization energy:

Example Mg

1st ionization energy = 735 KJ/mol

2nd ionization energy = 1445 KJ/mol

3rd ionization energy = 7730 KJ/mol

The big gap between 2nd and 3rd I.E shows that 3rd electron is being removed from the inner shell and Mg has only two electrons in its outermost shell.

Successive Ionization Energies in Kilojoules per Mole for the Elements in Period 3

General increase →								
	Element	I_1	I_2	I_3	I_4	I_5	I_6	I_7
↑ General decrease	Na	495	4560					
	Mg	735	1445	7730	Core electrons*			
	Al	580	1815	2740	11,600			
	Si	780	1575	3220	4350	16,100		
	P	1060	1890	2905	4950	6270	21,200	
	S	1005	2260	3375	4565	6950	8490	27,000
	Cl	1255	2295	3850	5160	6560	9360	11,000
	Ar	1527	2665	3945	5770	7230	8780	12,000

2.7: Mass spectrometry

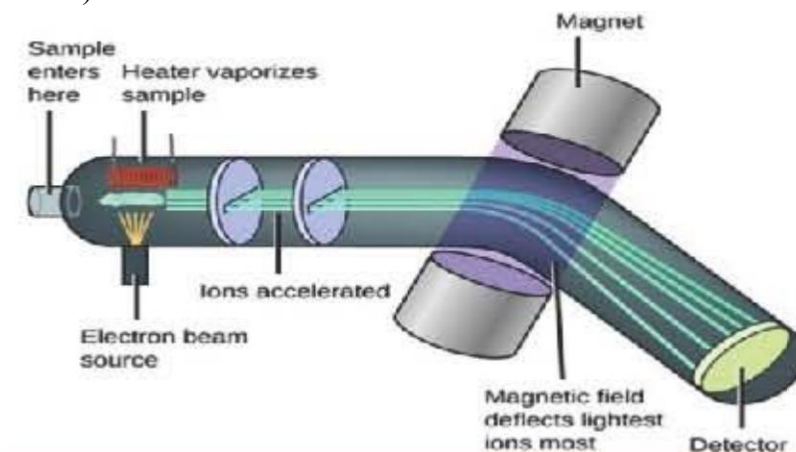
This is an analytical technique that is used to measure;

- ✓ Exact masses of isotopes of an element.
- ✓ Relative abundance of isotopes.
- ✓ Molecular masses of compound.
- ✓ Molecular structure

Principle

"In this technique a substance is first volatilized (vaporized) then ionized with the help of high energy beam of electrons. The gaseous positive ions thus obtained are separated on the basis of their M/z value and recorded in the form of peaks."

- ✓ Vaporization
- ✓ Ionization
- ✓ Acceleration
- ✓ Deflection
- ✓ Recording (collection)



Working

Ionization:

The process involves the use of laser to vaporize the sample and then bombarding it with high-energy electrons introducing to a high voltage field to produce ions.

Ion separation:

The ions are accelerated and directed into mass analyzer.

Mass analyzer separate the ions on the basis of m/z .

It allows the different ions to focused and detected at different times.

Detection:

For exiting the ions from mass analyzer, they hit the detector.

The detector records the times for each ion to reach it.

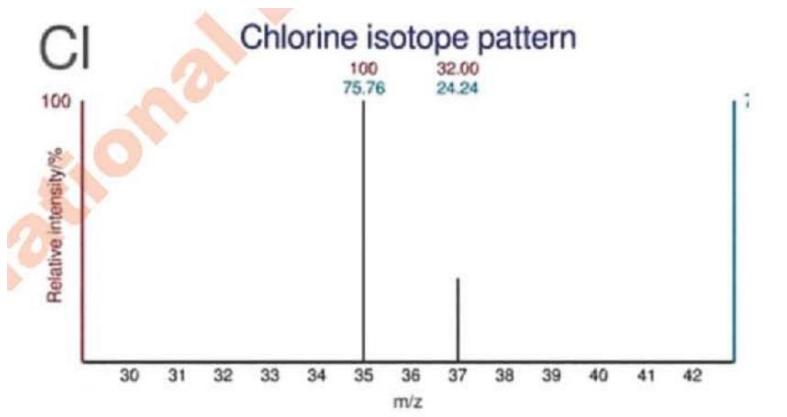
This data is used to calculate the m/z for each ion

Data Analysis/ Mass spectrum

A plot of data in such a way that m/z value of isotope of an element are taken at x-axis (abscissa) and relative abundance of ion on y-axis.

Number of peaks tell about the number of isotopes.

Position of peak tell about the stability of abundance of isotope.



Mass spectrometer / Spectrograph

“An instrument which is used to measure exact masses of isotopes of an element along with their relative abundance.”

Types

1. Aston’s mass spectrograph:

Discovered by Aston (1919)

Used to identify the isotopes of an element on the basis of their atomic masses.

Aston was awarded Nobel prize in 1922 for the discovery of isotopes by means of mass spectrograph

2. Dempster’ mass spectrometer:

A.J Dempster in 1918

This instrument is used for the identification of elements which were available in solid state.

Modern mass spectrograph

DMS+ amplifier+recorder

Applications of Mass spectrometry

- ✓ Mass spectrometer is used to:
- ✓ Identify the relative abundance of isotopes of an element.
- ✓ To identify the unknown compounds.
- ✓ Quantify the amount of specific substance.
- ✓ Elucidating the molecular structure.
- ✓ Studying the biomolecules like nucleic acid and protein.

Calculating the atomic mass of an element

Following formula is used to calculate the average atomic mass of an element.

$$\text{Average atomic mass} = \frac{(\text{Mass No} \times \text{Relative abundance}) + (\text{Mass No} \times \text{Relative abundance})}{100}$$

Non-integer relative atomic mass

The non-integer atomic mass in the periodic table is the result of relative abundance of different isotopes.

Relative abundance is usually taken in %age unit.

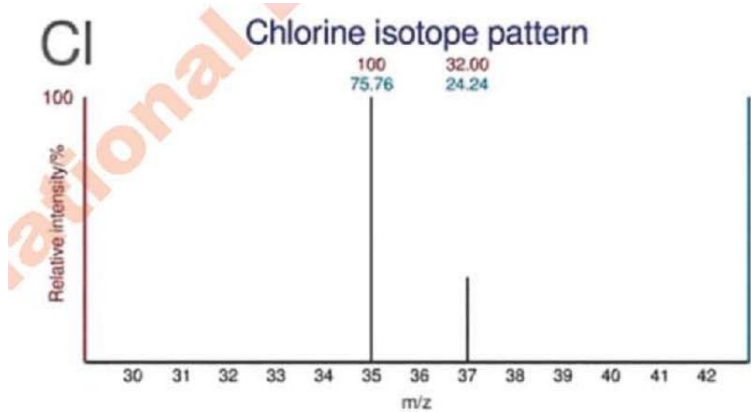
For example:

The diagram shows two isotopes of Cl.

Cl-35 (R.A=75.76% and Cl-37(R.A=24.24%)

$$\text{Average atomic mass} = \frac{(35 \times 75.76) + (37 \times 24.24)}{100}$$

Average atomic mass= 35.48 amu

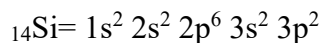


Electronic configuration of modern electronic material

Semi-conductor

“The materials having conductivity between the conductor and insulator are called semi-conductor.”

Electronic configuration of Si

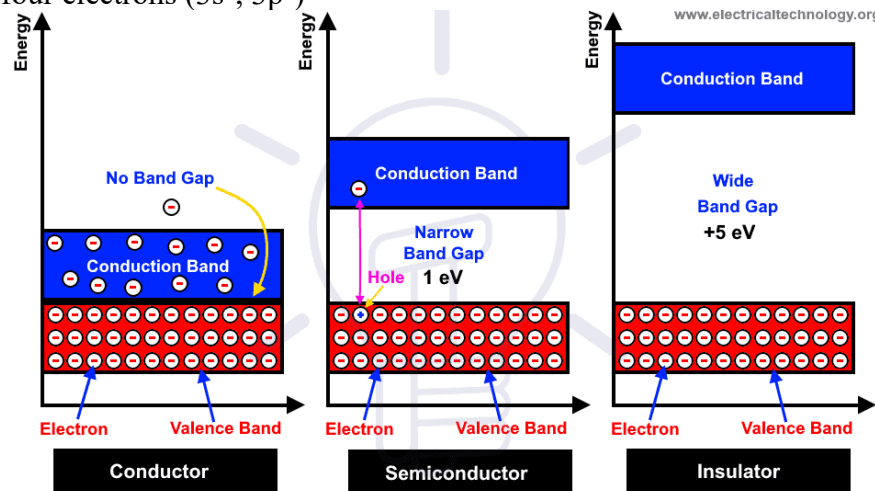


(k=2) (L=8) (M=4)

- 1st energy level (K)= 2 electrons = $1s^2$
- 2nd energy level (L)= 8 electron = $2s^2 2p^6$
- 3rd energy level (M)= 4 electrons= $3s^2 3p^2$

Role of electronic configuration in determination of type of semi-conductor

- ✓ 3rd M-shell is the valence shell of Si.
- ✓ It has total four electrons ($3s^2, 3p^2$)

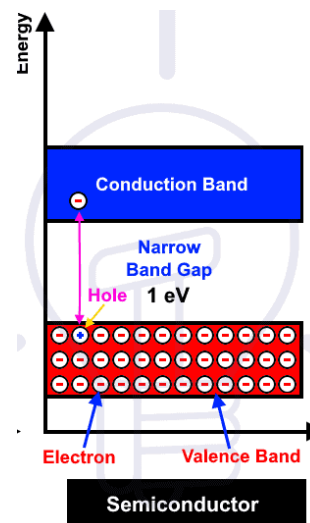
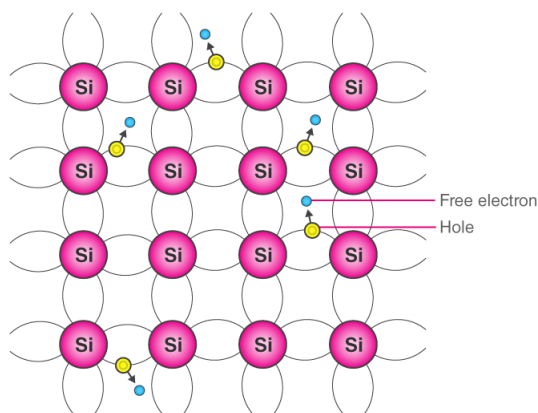


For silicon

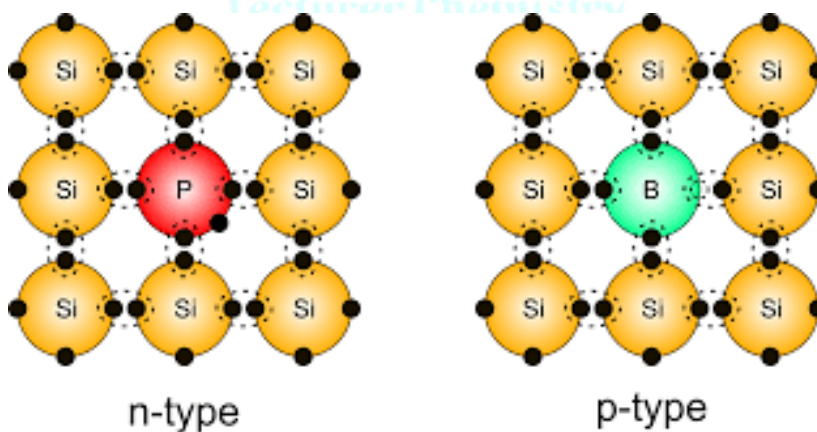
- ✓ In valence bond, electrons are tightly bond and cannot move freely.
- ✓ In ground state, there are non electrons in the conduction band
- ✓ Small energy gap.
- ✓ Electron can be promoted from valance band to conduction band creating
- ✓ electron-hole pair. This is called *excitation*.

Types of semi-conductor**1. Intrinsic semi-conductor:**

Pure semi-conductor (Si and Ge) are called intrinsic semi-conductor.

INTRINSIC SEMICONDUCTORSBYJU'S
The Learning App**Extrinsic semi-conductor:**

Impure semi-conductor (Doped with n-types which introduce extra electron or p-type material which have hole and behave like positive charge carrier extrinsic semi-conductor).

**Doping:**

A process in which some silicon atom is substituted with some other element like phosphorus (for n-type) and Boron (p-type) is called doping.

Types of Extrinsic Semi-conductor

- ✓ Electronic configuration of Si ,
- ✓ $_{14}\text{Si} = 1s^2 2s^2 2p^6 3s^2 3p^2$
- ✓ This special electronic configuration make it either accept or lose the electron.

n-type semiconductor:

- ✓ When silicon is doped with P or N atom it is called n-type semi conductor.
- ✓ Dopants introduce the excess electron that can move freely in the conduction band, improving the electrical conductivity.

p-type semiconductor:

- ✓ When silicon is doped with B it is called p-type semi-conductor.
- ✓ Dopant create a hole in valence band, can move behaving as positive charge carrier.

Applications of Semi-conductor

By using n-type and p-type materials engineers can make;

- ✓ Diodes
- ✓ Resistors
- ✓ Other electronic components that control the flow of electrical current.
- ✓ Semi-conductors are the basis of modern electronics and technology.

For More Information:

Online and physical tuition is available

Muhammad Amjad

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